



LUCIE-3D: A three-dimensional climate emulator for forced responses

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Introduction

We introduce LUCIE-3D, a lightweight three-dimensional climate emulator designed to capture the vertical structure of the atmosphere, respond to climate change forcings, and maintain computational efficiency with long-term stability. Building on the original LUCIE-2D framework, LUCIE-3D employs a Spherical Fourier Neural Operator (SFNO) backbone and is trained on 30 years of ERA5 reanalysis data. The model incorporates atmospheric CO₂ as a forcing variable and optionally integrates prescribed sea surface temperature (SST) to simulate coupled ocean–atmosphere dynamics.

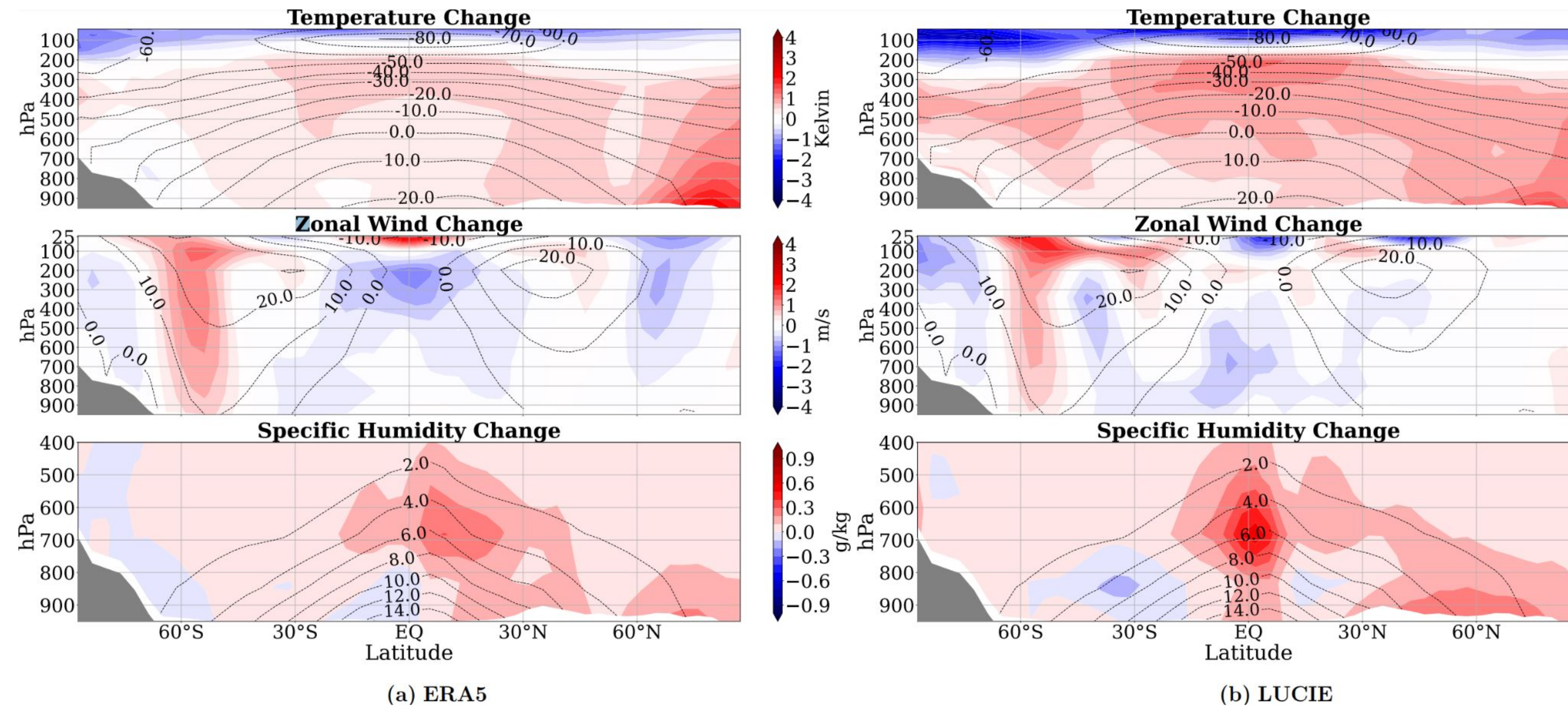


Figure 1: Zonal climate change map of (a) ERA5 (b) LUCIE-3D.

Warming Trend

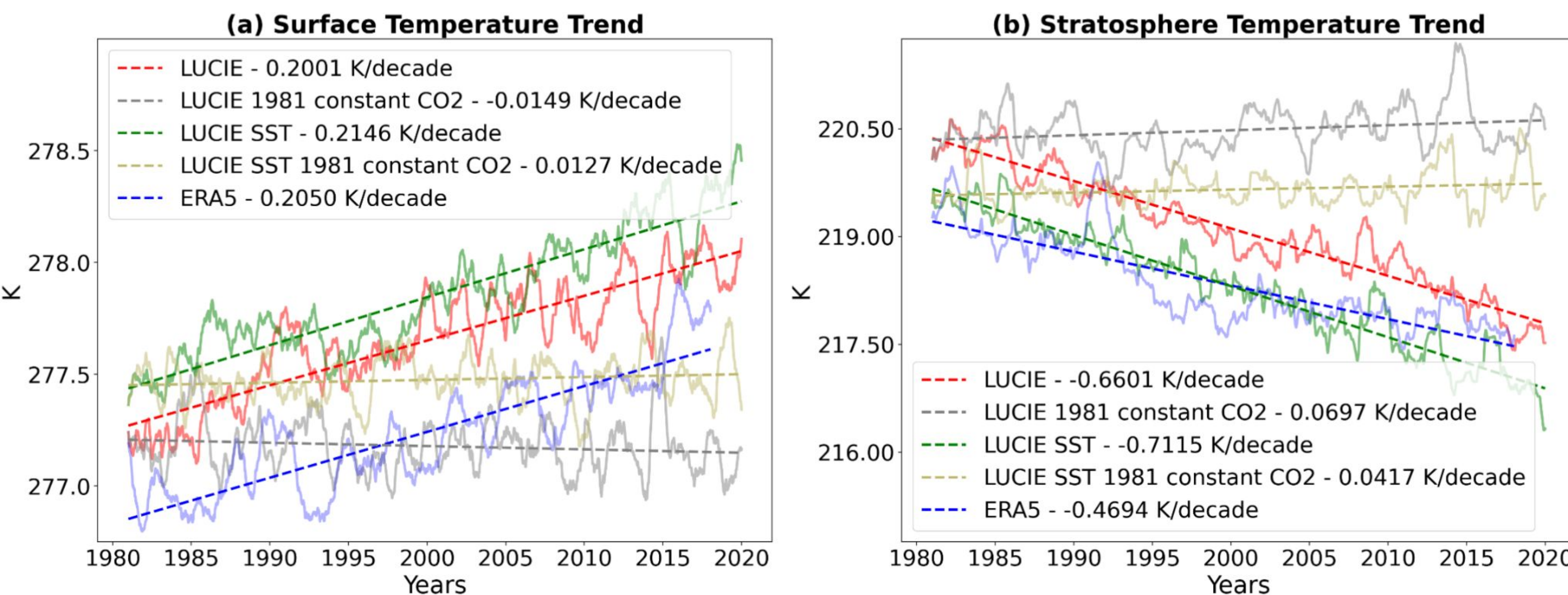


Figure 2: Global averaged warming trend of ERA5. a) surface temperature and (b) stratosphere temperature. LUCIE-3D models are separated into LUCIE-3D and LUCIE-3D trained with SST forcing. Both models are run in inference mode with real CO₂ (red and green lines) and stationary CO₂ (grey and yellow lines). The blue lines represent ERA5.

Climatology spin-up

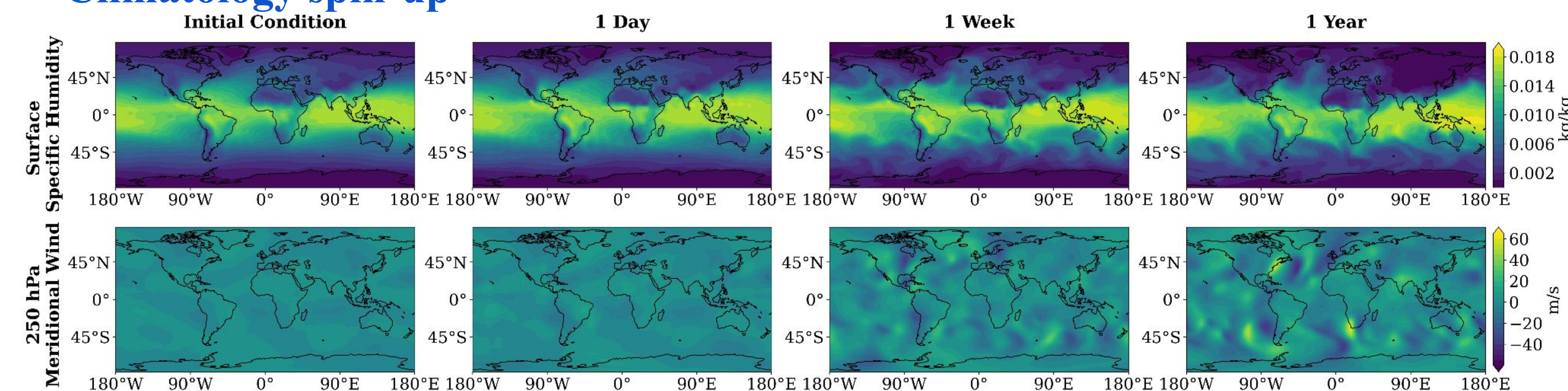


Figure 6: LUCIE-3D initialized with climatology of 1981-2000.

Precipitation

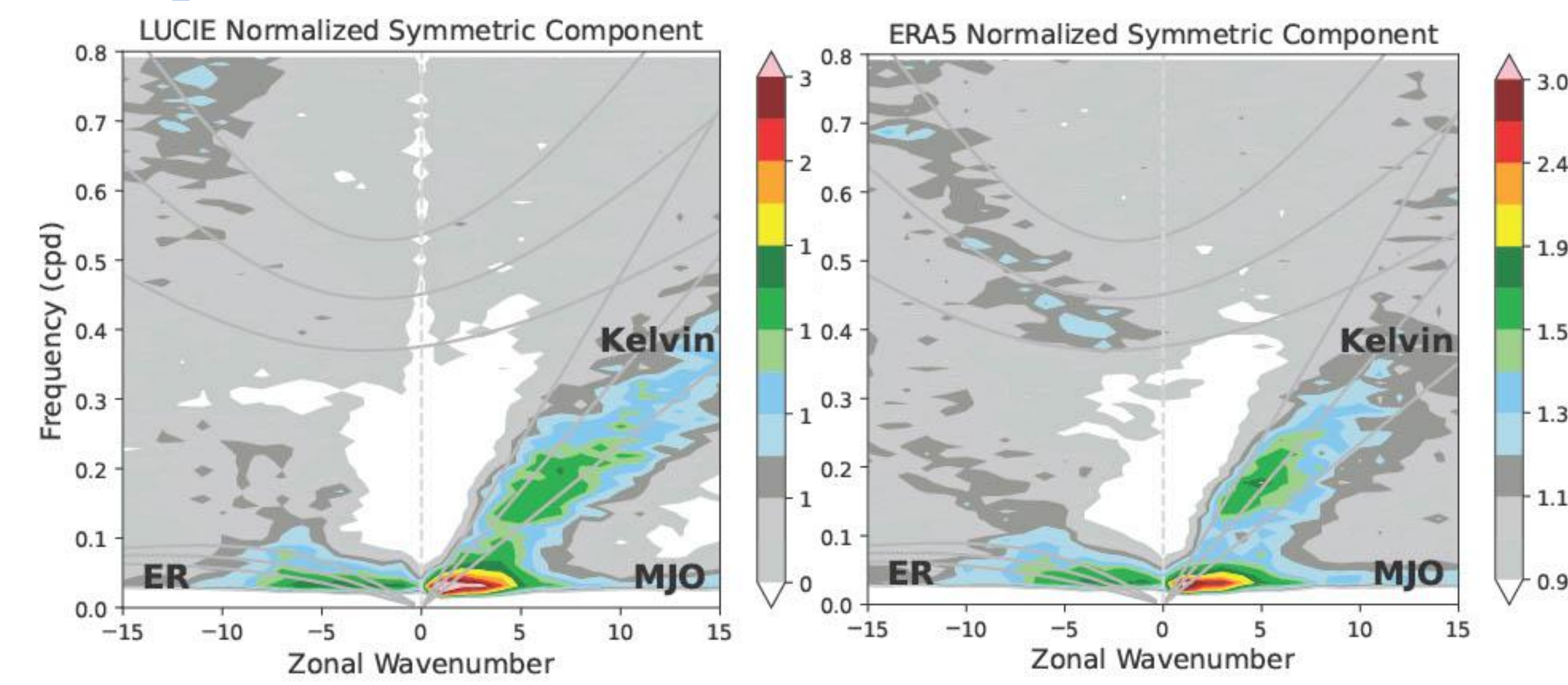


Figure 3: Wheeler-Kiladis diagram of LUCIE-3D and ERA5. Horizontal axis represents the zonal wavenumber ranging from -15 (westward) to +15 (eastward) and the vertical axis represents the frequency. The shading represents the spectral power with long-term climatology. The gray contour lines trace theoretical dispersion relations for equatorially trapped wave modes.

Sudden Stratospheric Warming

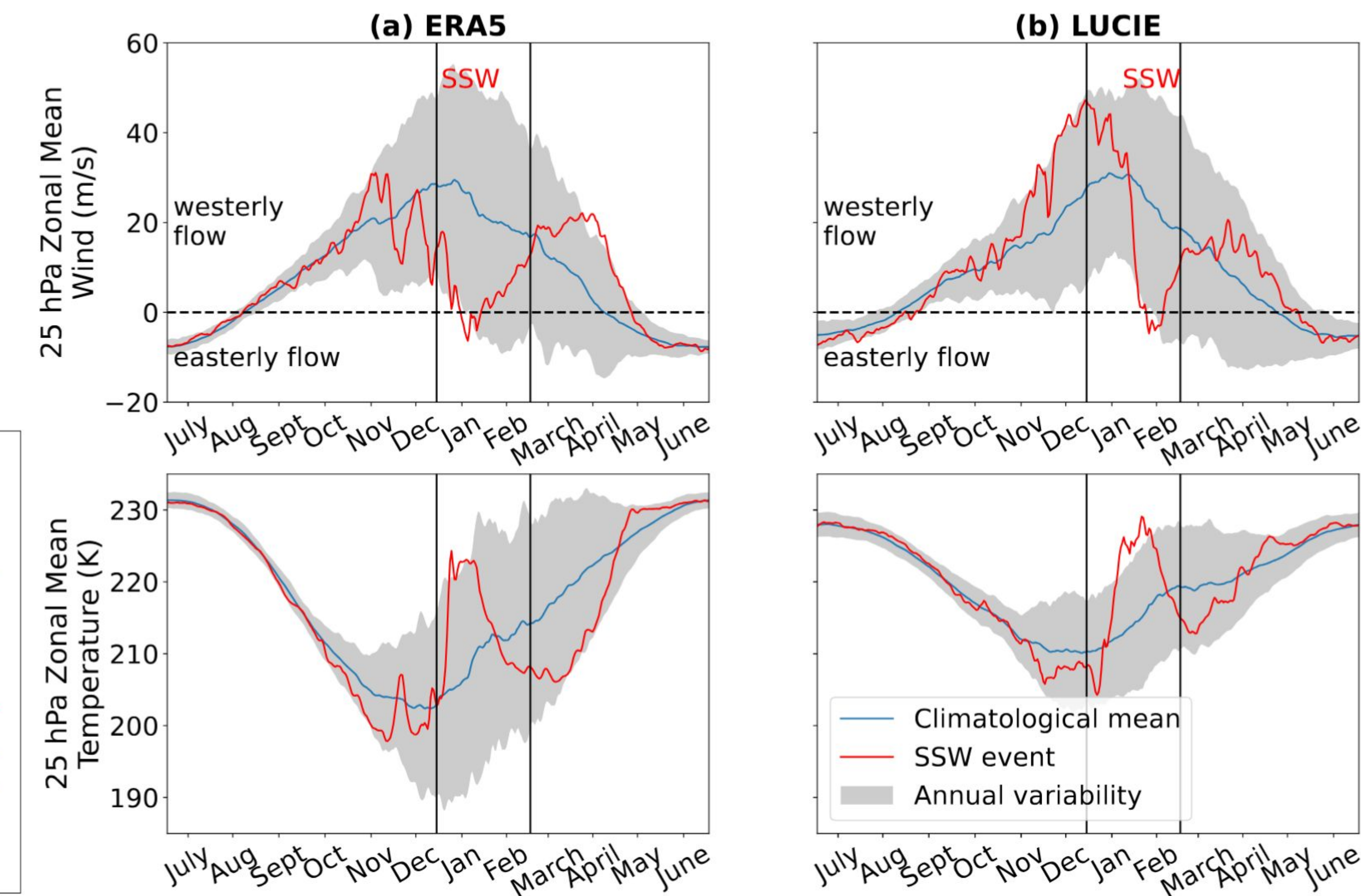


Figure 4: The performance of LUCIE-3D in capturing SSW. Results are shown at the top model level (25 hPa pressure level) for (top panels) the mean of the zonal wind component in the 55°N–65°N latitude band, and (bottom panels) the mean temperature north of 60°N. Blue curves show the climatological daily mean, while the gray shading characterizes the annual variability by displaying the range between plus and minus two standard deviations. The ERA5 SSW event happened in 2013. The LUCIE captured SSW event happened in 2016.

Zero spin-up

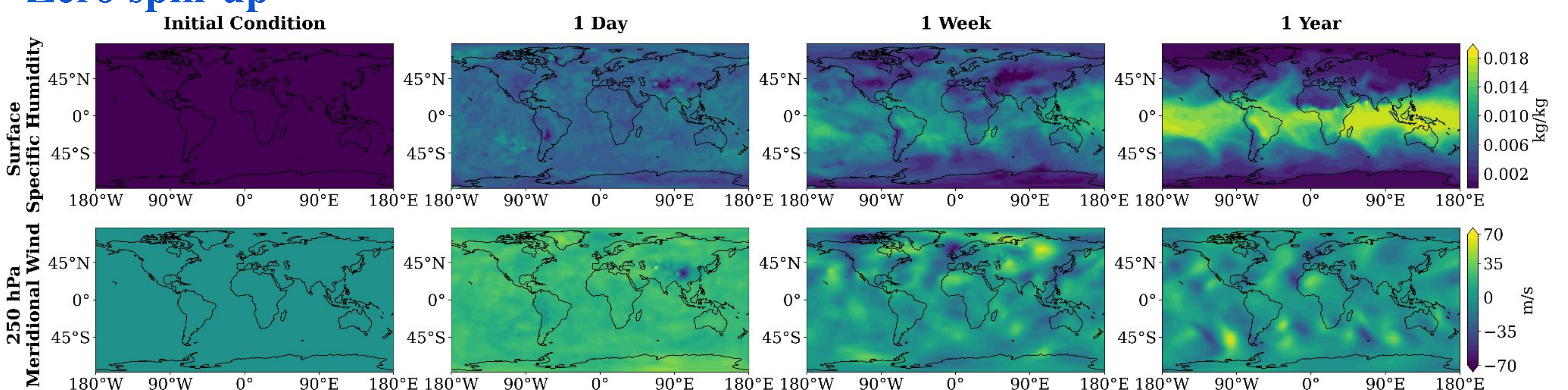


Figure 7: LUCIE-3D initialized with "zero atmosphere".

Figure 6 shows the model perform autoregressive inference with LUCIE-3D by initializing the model with a 30-year climatological mean of the prognostic variables, while using the actual historical forcing variables at each timestep. Figure 7 tests the model under more extreme conditions, we conduct an experiment in which LUCIE-3D is initialized with a near-zero values for all prognostic variables while retaining accurate external forcings. Both inferences converged to realistic climatological states.